

Direct Anterior Restorations

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As esthetic awareness grows and becomes more important throughout our society, patients demand highly esthetic restorations. Esthetic restorative materials must blend into the natural dentition by simulating the natural tooth in color, translucence, form, and texture yet also have adequate strength and wear characteristics, good marginal adaptation and sealing, insolubility, and biocompatibility. These materials must also remain color stable and maintain external tooth morphology to provide a functional, lasting esthetic restoration.[1] This chapter addresses the materials and clinical procedures used to place direct functional esthetic restorations in anterior teeth.

Resin composites, also called *resin-based composites*, *composite resins*, or simply *composites*, are by far the most commonly used restorative materials in the anterior region of the mouth.[2] Resin composites are currently the direct restorative materials that best fulfill the requirements of tooth preservation, excellent esthetics,

and durability. Resin composites may be used for the treatment of caries and tooth defects as well as for solving esthetic problems like diastemata, peg-shaped lateral incisors, misaligned teeth, and discolorations. Nonetheless, resin composites present some disadvantages: *(1)* they shrink during polymerization, *(2)* they require good isolation for proper bonding, *(3)* they are less resistant to wear than ceramics, *(4)* they are prone to chipping, *(5)* they do not bond as well to dentin and root surfaces as they do to enamel, and *(6)* they do not have antibacterial properties.[3]

Studies assessing the clinical longevity of resin composite as an anterior restorative material have found that 60% to 80% of all Class 3 and Class 5 restorations remain acceptable after 5 years. The most common reasons for repair or replacement of anterior composite restorations are marginal staining, surface discoloration, secondary caries, edge chipping in high-stress areas, and/or fracture of the restoration.[4–9] Class 4 composite restorations placed on fractured teeth present shorter longevity than Class 3 or Class 5 restorations because they are submitted to more stress from occlusal function.[10,11] The most common variables influencing the survival of anterior composite restorations are reported to be the individual practitioner, the location and size of the restoration, the patient age and caries risk, and the bonding substrate.[12–14]

Material Considerations

Resin composites

Resin composites contain four structural components: polymer matrix, filler particles, a coupling agent, and an initiator system. The *matrix* is the continuous phase to which the other ingredients are added. Most resin composite matrices are based on the bisphenol glycidyl methacrylate (bis-GMA) resin developed by R. L. Bowen of the National Institute of Standards and Technology and patented in 1962. Some resin composites use urethane dimethacrylate (UDMA) instead of bis-GMA, while many now use a combination of the two materials. Because of the high viscosity of bis-GMA, manufacturers typically add a low-viscosity monomer, such as triethylene glycol dimethacrylate (TEGDMA), as a polymerizable diluent. Recent progress in resin composite formulation includes the development of new monomers with reduced polymerization shrinkage or shrinkage stress and the incorporation of adhesive monomers to make self-adhesive composites.[3]

Filler particles are usually a type of radiopaque glass, zirconium oxide,

aluminum oxide, or silicon dioxide added to the matrix to improve the physical properties of the final composite. The filler improves translucency; reduces the coefficient of thermal expansion; reduces polymerization shrinkage of the composite; and makes the material stronger, harder, denser, and more resistant to wear. Generally, the greater the percentage of filler added (by volume or weight), the better the physical properties of the resin composite. However, filler loading has an upper limit, after which the material becomes too viscous for clinical use.

The filler particles are coated with an *organosilane*, a coupling agent capable of producing chemical bonding to the filler particles and the resin matrix. Without a coupling agent, the strength of the cohesive mass is reduced, and the filler particles tend to be lost, or "plucked," from the surface as preferential wear occurs in the softer surrounding resin matrix.[15]

Activation of the curing reaction may be accomplished by mixing two composite components (ie, autocuring or self-curing composites) or by exposing a single component to light of the proper wavelength (ie, light-cured composite). For self-curing materials, the first type of composites developed, one composite paste contains an initiator, typically benzoyl peroxide, which when attached by an amine molecule from the second paste splits into two active free radical molecules capable of initiating the polymerization reaction of methacrylate monomers. The more popular light-cured composites require a photoinitiator to activate the polymerization reaction, with camphorquinone being the most commonly used molecule. Current resin composite restorative materials require exposure to visible light, with most of the energy delivered in the range of 460 to 480 nm (blue light), which is the portion of the visible spectrum in which camphorquinone is most sensitive. The photoinitiation is most efficient when an amine molecule is also included in the formulation to act as a type of accelerator in the presence of the camphorquinone.

Light-cured resin composites are packaged in syringes or in unit doses. The main advantage of purchasing material in syringes is their lower cost. Unit doses allow insertion of the material directly into the preparation, minimizing trapped air bubbles.[16] The unit doses also make infection control procedures easier because they are discarded after use and require no disinfection. The use of syringes requires special attention during dispensing and handling. When dispensing, the material must be extruded in adequate amount and cut with a bladed instrument to avoid contamination from abrading or cutting the tip of the plastic syringe. Because the syringe is typically black to keep the material protected from external light, which would cause it to prematurely cure, small particles from the plastic syringe would

lead to a discoloring of the composite.

Physical characteristics

Resin composites have steadily improved through the years; they are now durable, esthetic, and predictable. Used in combination with an adhesive system, resin composites are reliably and durably bonded to dentin and enamel. Although adhesion to dentin is not yet as reliable as adhesion to enamel, advances in the understanding of dentin microstructure, permeability, and the bonded interface have steadily improved the quality and success of adhesive systems in recent years (see chapter 9).

Resin composites have several undesirable characteristics that must be overcome to achieve long-term clinical success. Volumetric shrinkage during polymerization is typically between 1.5% and 5.0%, which, when the material is placed into a bonded cavity preparation, generates internal stresses in the composite that create additional stresses at the bonded tooth-composite interface, potentially leading to marginal defects.[17,18] Shrinkage stresses that occur in the early phase of polymerization when the composite is still relatively fluid are effectively relieved by deformation and flow of the material.[19] However, stresses occurring later in the process, after the material has acquired significant rigidity (called *gelation*), are not relieved by material flow. These residual stresses may leave the composite weakened and may reduce the adhesion to the tooth.[20,21] These stresses also may cause gap formation at the cavosurface margins, especially at those with the weakest bonds (usually dentin or cementum).[22] Marginal gaps may result in microleakage, sensitivity, staining at the margins of the restoration, and recurrent caries.[23–25] Investigations have determined that nanoleakage at the resin-dentin interface, as well as hydrolytic and enzymatic degradation of collagen bonds, contribute to a weakened adhesive interface over time.[26–28]

The use of incremental placement techniques, long enamel bevels, low-shrinkage materials, flexible resin liners, slow-setting resin-modified glass-ionomer liners, and modified light-curing protocols have all been recommended to help offset the effects of polymerization shrinkage and its stresses.[29–39] The *configuration factor* (*C-factor*) of a restoration, which is defined as the ratio of the bonded surface area to the nonbonded surface area of the restoration, influences the intensity of the stresses produced during polymerization. High C-factor preparations are conducive to high shrinkage stress compared with low C-factor preparations, because more of the composite is constrained and not free to flow or deform to relax stresses.[20,40]

Fortunately, for most anterior restorations the C-factor is favorable and stress at the adhesive interface is minimal. Additionally, the viscoelastic properties of the material influence shrinkage stress. High-stiffness materials will typically generate higher shrinkage stress, increasing the possibility of gap formation at the margins.

Resin composites have a coefficient of thermal expansion two to six times higher than that of tooth structure.[41] This means that resin composite expands and contracts more and at a greater rate than does tooth structure in response to changes in temperature. This mismatch may contribute to loss of adhesion and greater microleakage.[25] Over the years, significant improvements in adhesive systems have helped offset some of the inherent problems associated with resin composites.

Handling characteristics

An important factor for the clinician in the selection of a resin composite is the handling of the material. Ideally, resin composite materials should be soft and easily manipulated, should not stick to placement instruments and brushes, should adapt well to the cavity walls, and should not slump during placement. Handling is greatly influenced by the viscosity of the resin composite. According to their viscosity, resin composites can be classified as *conventional*, *packable*, or *flowable*. Low-viscosity materials adapt well to cavity walls, but they tend to be sticky and are prone to slump. When the viscosity is extremely low, as is the case with flowable resin composites, the risk of porosities inside the restoration increases if the material is not handled appropriately.[42] On the contrary, high-viscosity materials do not adapt to the tooth as easily and require careful attention during placement. Their use is advantageous because they are not sticky and maintain their shape during placement. Variation in the viscosity of resin composites does not always correlate directly with the filler content[43,44]; however, low-viscosity materials generally have lower filler loading. Differences in handling characteristics are also observed with variations in resin matrix chemistry, temperature, and humidity.[45–47]

Many instruments, offered in various shapes and surface coatings, are available for placing and shaping resin composite materials. The composition of a typical instrument kit contains a thin, flexible, bladed instrument; a small ovoid-shaped burnisher; a thin spatula; a concave-convex spatula; and different shapes of brushes (Fig 10-1). A gauze pad lightly moistened with alcohol may be used to clean the active part of the instrument during resin composite placement to prevent the material from sticking to the instrument. However, the alcohol should be allowed to evaporate from the instruments before contact with the resin material to avoid incorporating

alcohol into the composite. When using very sticky materials, it is recommended to use very small amounts of a purposely manufactured wetting resin to lubricate instruments (Fig 10-2). The use of these low-viscosity resins as lubricants must be tightly controlled because these materials could potentially dilute the resin composite and bring about a reduction in physical and mechanical properties.[48]



Fig 10-1 Commonly employed resin composite placement and contouring instruments and brushes.



Fig 10-2 Examples of wetting resins available on the market. They can be used in small amounts to lubricate instruments for easier manipulation of resin materials.

Optical characteristics

Modern resin composite kits sold by dental manufacturers contain multiple shades and various opacities for the purpose of matching shade and translucency of both enamel and dentin. This has facilitated the achievement of natural esthetics with resin composites. Unfortunately, there is great variability among materials and no consensus among manufacturers regarding nomenclature, shading, or degree of translucency/opacity.

The combination of optical properties from enamel and dentin provides the final appearance of normal tooth structure. These properties are modified by factors such as age, thickness of tooth tissue, and degree and quality of calcification. In a

"normal" unworn dentition, dentin provides chroma, opacity, and fluorescence. Enamel modifies the appearance of dentin by providing translucency and opalescence[49] (Fig 10-3).



Fig 10-3 Illustration emphasizing the differences in translucency for dentin *(left)* and enamel (overlay) *(right)* resin composite compositions. Applying layers with differing chroma and translucency and providing appropriate surface character can greatly enhance the lifelike appearance of anterior restorations.

Based on their optical properties, resin composite kits usually contain three types of materials: *(1)* a dentinlike material, designed to imitate the dentin's optical properties; *(2)* an enamel-like material, designed to simulate the enamel's optical properties; and *(3)* a translucent-like material, designed to mimic translucent areas of teeth. Manufacturers have various names for each type of material. Dentinlike materials may be called *opaque*, *dentin*, *body*, etc; enamel-like materials may be called *enamel*, *body*, etc; and translucent materials may be called *incisal*, *translucent*, etc. These different designations and names create confusion among practitioners. For a clear understanding of how to use a specific material, it is important for dentists to know what each name represents and the intended purpose of each system component.

When a single shade is used for a restoration, an intermediate opacity should be selected. Usually the manufacturer has designated the enamel-like material as intermediate opacity for this purpose. When multiple shades of resin composite are layered, the principle of replacing dentin with dentinlike materials and enamel with enamel-like materials should be observed (Fig 10-4).



Fig 10-4 Resin composite increments of the same thickness presenting four opacities: Dentin, Body, Enamel, and Translucent (Filtek Supreme Ultra, 3M ESPE).

Microfilled resin composites

Microfilled resin composites contain silica fillers of submicron size only. Because these small particles, with an average size of 0.04 μm, have a very high surface area–to-volume ratio, they require a large amount of monomer to wet their large surface area and therefore can only be incorporated into resin up to a relatively low volume fraction. Because the properties of the cured resin composite are dependent on the quantity of the stronger filler particle, these microfilled composites are relatively weak. To enhance the properties of these materials, manufacturers incorporate what are called *prepolymerized resin fillers* (PPRF) in addition to the submicron-sized fillers. The PPRF are produced by adding a high concentration of submicron fillers into a very dilute resin, then heat curing this viscous mixture to maximize polymerization, and finally grinding the filled polymer into 5- to 50-μm particles. These particles are then incorporated with additional microfiller particles to form the final restorative material for clinical use (Fig 10-5a). In this way, filler content is maximized, polymerization shrinkage is minimized, and the resin composite remains highly polishable because every individual silica particle is only about 0.04 μm in size.[50]

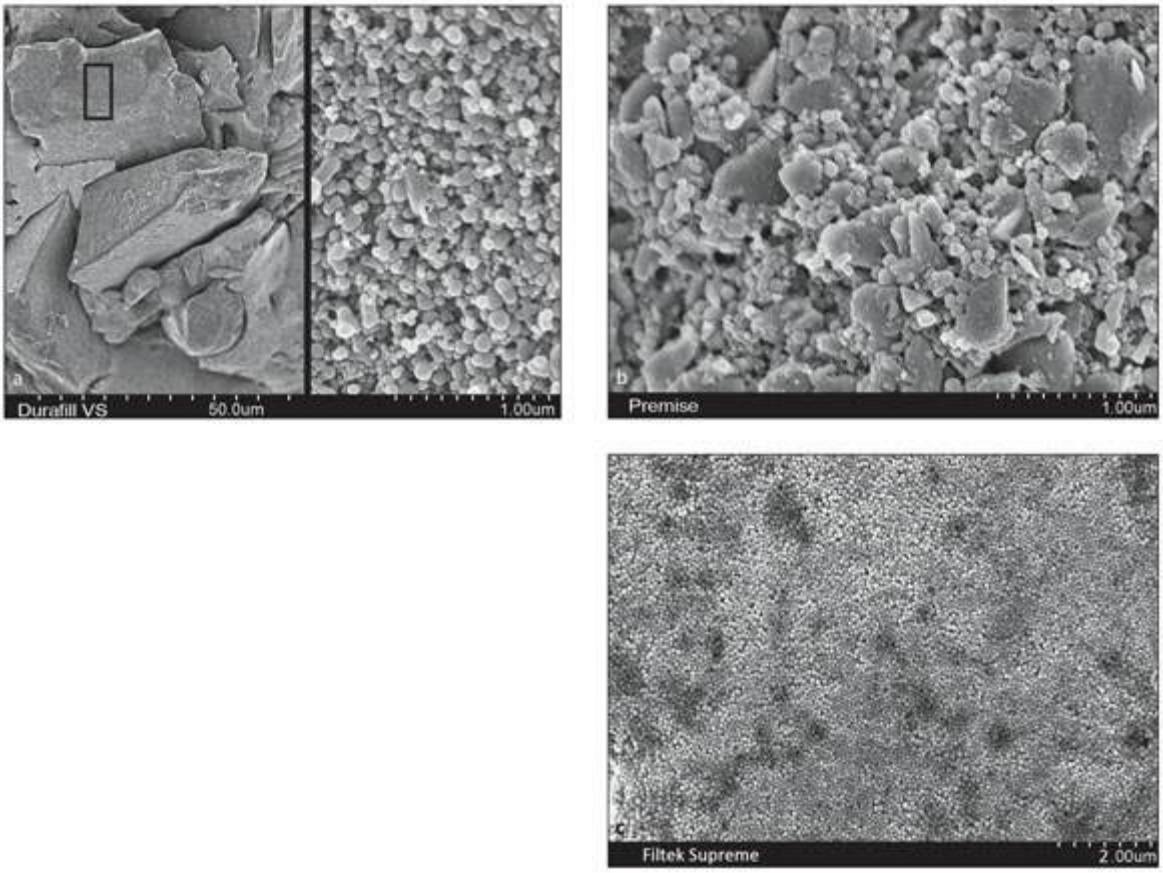


Fig 10-5 Most commonly used categories of resin composite: *(a)* the microfilled resins, such as Durafill VS (Heraeus Kulzer), contain only submicron particles *(right side)* along with PPRF *(left side)*; *(b)* the hybrid materials, such as Premise (Kerr), contain a combination of submicron particles and particles up to 4 µm in diameter; and *(c)* the nanofilled materials, such as Filtek Supreme Ultra, contain loose nano-sized filler particles (20 to 70 nm) and clustered nanosized particles.

Microfilled resin composites can be polished to the highest luster and smoothest surface of all the resin composites, and their primary indication is for esthetic areas where this luster is required, such as for direct resin composite veneers. Microfilled resin composites, in general, are not as strong as other types of composites and are not recommended for stress-bearing restorations.[41,51–54] Microfilled materials tend to absorb more water because of their increased resin content, with a resultant decrease in long-term color stability.[55] When a highly polished Class 4 restoration is needed, a hybrid material may be used as a substructure to maximize strength and wear resistance, then veneered with a microfilled resin composite for a smooth surface (see Fig 10-6).

Hybrid resin composites

As the name implies, *hybrid resin composites* contain a blend of submicron (0.04-µm) and small-particle (0.4- to 4.0-µm) fillers (Fig 10-5b). This combination of filler particles allows the highest levels of filler loading among current resin composites and a corresponding improvement in physical properties as compared with microfilled composites.[56,57] They can be polished to a fairly high luster, but not to the extent of a microfilled material. Hybrid resin composites are a combination of conventional and microfilled technology and can be used for anterior and posterior restorations. The high filler content also improves the hybrid material's resistance to internal discoloration.

In the last decade, dental manufacturers have fabricated hybrid resin composites with an average particle size of less than 1 µm. Because of the submicron size, these materials are called *microhybrids* (eg, Point 4, Kerr). Submicron particles can be measured in µm or nm (1 µm = 1,000 nm), which explains why some materials in this category have been called *nano-hybrids*. While not always the case, many nanohybrids also contain PPRF to reduce overall curing shrinkage. These microhybrids and nanohybrids are easier to manipulate and polish than the regular hybrid composites and also maintain their luster better over time.[58]

Nanofilled resin composites

A new category of material, termed *nanofilled resin composites*, has appeared on the market in recent years. The research and development of these materials were designed with the objective of successfully integrating a high concentration of nanoparticles into resin composites to enhance mechanical properties, handling, and ease of polishing.[59,60] Unfortunately, the term *nano-* has been mistaken as only referring to particle size. Actually, *nano-* refers to the technology and manipulation of nanosized particles with the purpose of improving the final performance of the product. Few materials with a true nano-fill technology exist in the market at the moment (eg, Filtek Supreme Ultra, 3M ESPE) (Fig 10-5c). In general, these nanofilled composites are nonsticky and nonslumping.

The first generation of nanofilled materials was easy to polish but did not hold its luster over time and appeared dull to patients and dentists. They also tended to be too translucent and had somewhat of a gray appearance. Changes in their composition have led to improved materials that are more esthetic, very easy to polish, and better maintain their luster over time.[61]

Glass-ionomer restorative materials

Glass-ionomer restorative materials are not commonly used when esthetics is a major consideration in anterior restorations. They are often recommended for patients with high caries risk.[62,63] A review of clinical trials that studied the cariostatic effect of fluoride-releasing restorative materials concluded that there is no definite evidence for or against a treatment effect of inhibition of secondary caries by these materials.[64] However, studies involving patients at high risk for caries due to xerosto-mia showed a significant reduction in caries incidence adjacent to resin-modified glass-ionomer restorations, as compared with other restorative materials.[65,66] The traditional, chemically cured glass-ionomer materials are not highly esthetic, but esthetics has been greatly improved in the resin-modified glass-ionomer materials. From a physical property perspective, the resistance against erosive wear of both chemically cured and resin-modified glass-ionomer cements is lower than that for resin composites. Clinical trials of 5 years' duration have reported high retention rates in Class 5 restorations.[67–69] Though not as color stable nor as durable as resin composites, these materials are suitable for use in visible anterior areas when dentin margins are prevalent or when the patient has been identified as being at high risk for developing new caries lesions.[70] Glass-ionomer restorative materials are discussed in more depth in chapter 14.

Interproximal Lesions: Class 3

Interproximal caries lesions are smooth-surface lesions found slightly gingival to the proximal contact, without involving the incisal angle of the tooth. These lesions can usually be detected with an explorer, radiographically, or with transillumination. Clinical changes in translucency may be evident and may be enhanced if light is directed through the proximal area using a focused, intense light source (ie, *transillumination*). Caries lesions cause a more opaque appearance of tooth structure. Incipient lesions tend to be V-shaped and confined to enamel; deeper lesions tend to spread within dentin.

Incipient enamel caries lesions

A proximal noncavitated enamel lesion may not require restorative treatment. Although there is no doubt that the lesion is pathologic, research and clinical experience have shown that this lesion can be arrested or remineralized by

preventive therapies and regular topical fluoride application. Developing a caries-management plan based on individual risk factors will decrease the possibility that future invasive procedures will be necessary. Evidence supports the viability of remineralization of caries lesions in enamel.[71–75] Chapter 5 contains a more complete discussion on caries and remineralization processes.

Cavitated enamel caries lesions

When the enamel surface is cavitated, it is past the point of remineralization. If the cavitation is very shallow and deeper enamel has been remineralized, a restoration may not be necessary unless the lesion is esthetically displeasing. If the lesion is confined to enamel, enameloplasty or recontouring may be sufficient. If necessary, a conservative cavity preparation may be accomplished and then restored.

Dentinal caries lesions

With peripheral enamel margins

When the caries lesion has reached dentin, it then becomes necessary to remove caries with cavity preparation and restoration. When all margins are located in enamel, a combination of an adhesive and a resin composite are the restorative materials of choice.

With margins extending onto the root surface

In areas where there is little or no enamel for bonding, the marginal adaptation and seal of the restoration may be optimized in two distinct ways. For high-caries-risk patients, an open sandwich technique using a resin-modified glass-ionomer restorative material may be employed to seal the cervical portion of the restoration. The remaining cavity is then filled with resin composite for improved esthetics.[76] There is a debate regarding the effectiveness of the open sandwich technique, with studies showing both advantages and disadvantages of its use. However, studies demonstrate that the open sandwich technique reduces postoperative sensitivity and provides protection of dentin margins from demineralization.[77–79]

For low-caries-risk patients, the approach to treating lesions with dentinal margins is simply to employ a well-proven dentin bonding system and to restore with a resin composite. Because current dentin bonding systems have been demonstrated to be very effective in providing retention, mechanical undercuts at the

expense of healthy tooth structure have become unnecessary.[80]

Interproximal Lesions Involving the Incisal Angle: Class 4

Interproximal lesions involving the incisal angle are usually the result of undermining and compromising the integrity of the incisal enamel. This undermining can occur as a result of untreated large interproximal caries or when replacing a large Class 3 restoration. Additionally, these lesions can occur due to an injury that results in a fracture of the incisal angle. The need for Class 4 restorations due to traumatic fracture occurs most often among children or young adults. The frequency of fractures of permanent incisors in children is reported to range from 2.4%[81] to 20%.[82] Traumatic fractures are likely to be more horizontal than vertical.

Shade or Color Selection

Selection of the shade or color of resin composite restorative material is an important and sometimes demanding step in restoring an anterior tooth. Factors that influence shade selection, such as proper lighting, environmental factors, and color acuity and eye fatigue are also discussed in chapter 4.

Proper lighting

One of the first requirements for a good color match is proper lighting. Commonly used fluorescent tubes emit light with a green tint that can distort color perception. Color-corrected fluorescent tubes that approximate natural daylight are available and are recommended for dental treatment rooms. The objective is to obtain shadow-free, color-balanced illumination without distracting glare or false colors[83] (Box 10-1). Additionally, the operatory light should be turned off because the orange-yellow wavelengths of these lights will interfere with adequate shade selection. If proper lighting is not available, color selection can be made near a window. However, even daylight varies considerably from day to day and throughout the day. When shade selection is critical, it is wise to use multiple light sources to choose the best shade and to avoid problems with *metamerism*, a complication observed when the perceived color of objects (in this case, teeth and resin composite restorations) is different under different light sources.

Box 10-1	**Color-corrected lighting**[83]
Overhead lights (fluorescent tubes)	
• Color Rendering Index (CRI): 90 or higher • Spectral energy distribution (SED): Natural daylight • Color temperature: 5,500 K • Illumination intensity: Approximately 150 to 200 foot candles at 30 inches above floor	
Dental operating light	
• Illumination intensity: 1,000 to 2,000 foot-candles • Color temperature: Optimum 5,000 K, should be adjust able from 4,500 K to 5,500 K to assist in color matching	

Environmental factors

The environment influences the shade selection. To improve the accuracy of shade selection, the color of the room walls and of the patient and staff clothing should be neutral to avoid imparting a negative color cast. Additionally, the patient should be asked to remove lipstick prior to shade selection.

Color acuity and eye fatigue

When selecting color or shade, the operator should avoid staring at the tooth and shade guide for long periods of time. Staring at these objects during shade selection will cause the colors to blend, resulting in a loss of color acuity. The shade guide should be placed adjacent to the tooth to be restored and then viewed briefly to determine which shade or shades match the color of the tooth; then the eyes should be moved away. Arranging the shade guide based on the value or brightness (see Fig 4-10) of the shades can also facilitate shade selection. Ideally, the eyes should be "rested" by viewing the horizon through a window or by looking at an object that is a muted blue, violet, or gray color. Full details of shade matching are addressed in chapter 4.

The dental assistant and patient can also assist in shade selection. By viewing the shade guide and the tooth from several positions and by accepting input from the assistant and patient, the dentist can achieve an acceptable color match.

Achieving optimal color selection

With proper attention to detail, resin composites allow for a very predictable shade

blending. The color or shade selection should be accomplished before the restorative procedure is initiated, prior to rubber dam placement or cavity preparation. If patients have whitened their teeth, it is necessary to wait 2 weeks after bleaching for shade selection and optimal bonding results.[84] The tooth should be made clean and free of contaminants by using a prophylaxis cup with a slurry of pumice to remove plaque or debris from the tooth surface and to eliminate any stains. Teeth should be kept wet and hydrated during shade selection because dehydration causes *(1)* significant lightening of the color and *(2)* an increase in the opacity of enamel and dentin.

The patient should be positioned in such a way that the teeth receive enough light from the illuminant or light fixtures. Most manufacturers provide or recommend a shade guide for their products to offer an approximation of the colors available. The shade guides are only helpful for a general determination of the shade. Most of these shade guides are made of acrylic resins that have different optical properties than resin composites. Many composites are fabricated to match VITA shades, but some others use their own shade designations. Matching between the VITA shade guide and the corresponding shades of resin composites and glass ionomers has been shown to be relatively poor.[85–87] A major drawback of all the shade guides is the fact that the shade of the underlying tooth is not taken into account. There will always be an effect on the shade of the restoration caused by the underlying tooth structure unless the restoration is more than 2.5 mm thick.[88–90] To overcome problems associated with shade guide discrepancies, custom shade tabs should be fabricated for each shade of the resin composite material used in the practice. The shades should then be arranged by value or brightness, from light to dark (see Fig 4-10). These tabs are then held incisally adjacent to the tooth to be matched, and the selection can be narrowed down to three to four shades. For selection of the final shade, the middle portion of the tooth should be observed and matched to the closest of these shade tabs. The selected shade then becomes the overall or "basic shade" of the restoration (see Fig 4-19).

When a layering and multishaded technique is used to replicate missing tooth structure, it is important to realize that the thickness of each layer will affect the final shade. In esthetically challenging situations, the ultimate shade selection is best achieved by producing a mock-up using each of the different layers to accurately match not just a single shade but also the adjacent tooth structure.[91] The shade can be confirmed with a small amount of resin composite applied as a test shade, placed adjacent to the tooth and cured. This procedure should only be performed on unetched surfaces to facilitate removal after shade verification. For Class 4

restorations in which no tooth structure will remain lingual or facial to the planned restoration, the test shade should be placed in the approximate thickness of the tooth structure to be replaced to ensure adequate opacity or color density.

Tinting and opaquing

Many manufacturers of resin composites provide accessory shades that contain a number of intense colors and opaque resins, premixed in syringes or bottles. These materials are normally not necessary in conservative Class 3 restorations but can play an important role in large Class 4 restorations, diastema closures, and direct veneers. Opaque shades, or composites designated as dentin replacements, are used to block the show-through of darkness from the mouth that may cause a Class 4 restoration to appear too dark or low in value or too translucent (Fig 10-6). Opaque resins may also be needed to mask discolored tooth structure.[89,90]

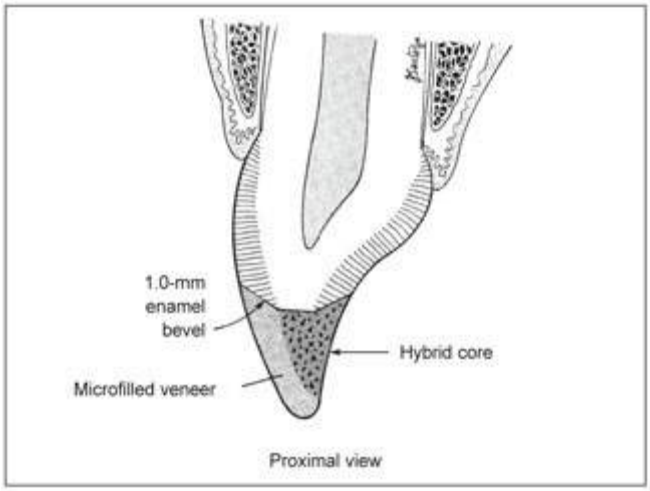


Fig 10-6 Opaque dentinlike resin composite should be used to block the show-through effect.

Masking agents should be applied in thin layers to allow sufficient space for the overlying composite to restore translucency. Use of the proper accessory shades can create the appearance of dentin overlaid with enamel. Accessory shades can also be used to recreate the yellow color seen in cervical areas or the translucency that appears in incisal areas. An ochre-shaded resin composite can be used to enhance chroma in the cervical areas, gray or blue may increase the "translucentincisal" effect, and white can be employed to reproduce a halo. Tints may be used to imitate white or brown spots that appear on adjacent teeth, although with the current ability to bleach teeth, the spots can usually be removed or their appearance neutralized with bleaching (see chapter 16).

Tooth Preparation

Class 3 interproximal lesions

Outline form for resin composite restorations is defect specific and determined solely by the extent of the caries lesion(s) and access for removal of carious tooth structure (Figs 10-7 to 10-9). There is no need for further extension of the preparation, and the removal of sound tooth structure to gain mechanical undercut retention is contraindicated.[92] Because the location of the caries in interproximal lesions is usually gingival to the contact point, the cavity preparation extends gingivally below the contact point. In contrast, the incisal portion of the contact point is not necessarily removed. When a lesion is limited to enamel, a round carbide or diamond bur is used in a high-speed handpiece for cavity preparation. The finished preparation resembles a saucer and has no retentive undercuts (see Fig 10-7). Adhesion to acid-etched enamel and dentin provides the necessary retention. Both laboratory and 3-year clinical data have demonstrated the durability of these saucer-shaped restorations.[93,94]

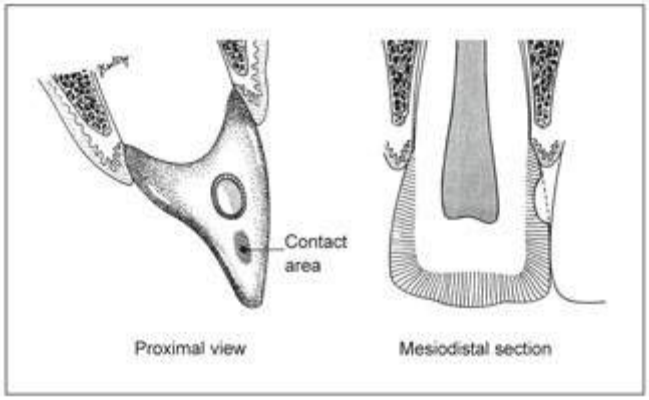


Fig 10-7 Saucer-shaped preparation in enamel. The caries lesion is usually located slightly gingival to the contact area. Every attempt should be made to maintain natural tooth contact between the adjacent teeth during restoration. If no cavitation is present, remineralization is preferable to restoration.

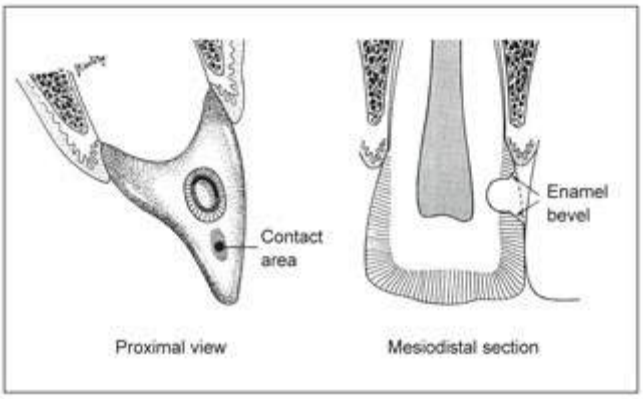


Fig 10-8 This preparation is similar to that in Fig 10-7, except the axial wall is extended into dentin and the external enamel margins are beveled.

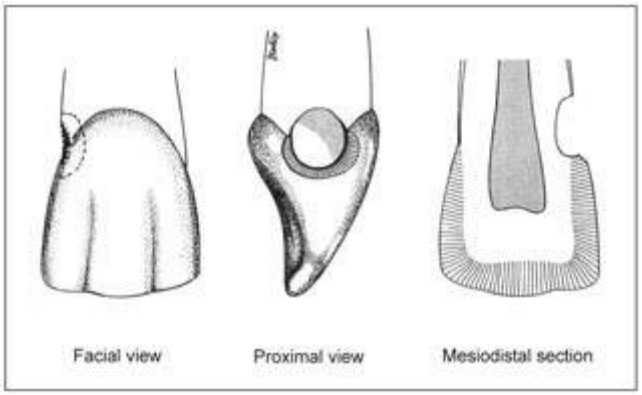


Fig 10-9 Caries lesion extending to the dentin-cementum surface. A bevel is placed only on enamel margins.

The lingual approach is preferred for larger Class 3 restorations, but it is not always possible depending on the location of the caries lesion. The number of burs used for cavity preparation should be kept to a minimum. A no. 2 round bur or a pear-shaped carbide bur or diamond in a high-speed handpiece can be used for initial access to the lesion. Initial penetration should be made through the marginal ridge, near the adjacent tooth but avoiding damage to it (Fig 10-10). The outline form of the preparation is then extended to provide access to the carious dentin. A spoon excavator and a large round bur may be used in the low-speed handpiece to excavate demineralized dentin.

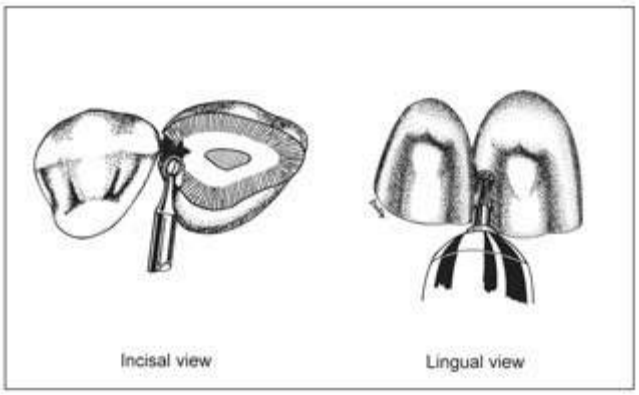


Fig 10-10 Initial penetration should be made through the marginal ridge, near the adjacent tooth but avoiding damage to it. Note the angulation of the long axis of the bur in relation to the location of the lesion.

The appearance of true enamel is more natural and esthetic than the most esthetic restorative material. To preserve facial esthetics, the Class 3 preparation should not be extended onto the facial surface unless necessitated by carious or missing enamel on that surface (Fig 10-11).

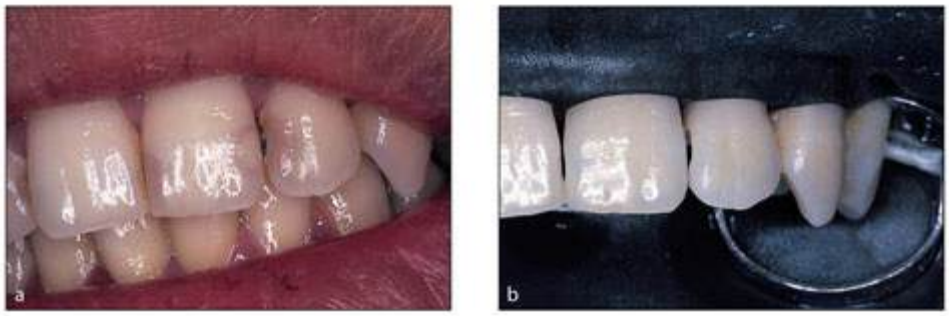


Fig 10-11 *(a)* Moderate-sized Class 3 caries lesions on the mesial and distal surfaces of the maxillary left central incisor and mesial surface of the lateral incisor. A preferable approach would be to leave facial enamel, if possible. *(b)* After cavity preparation, the labial margins are only slightly visible, and the labial enamel is supported by dentin.

Unsupported facial enamel may be left for internal etching and bonding to resin composite.[95] The facial approach for access to carious dentin is indicated only when the caries lesion already involves the facial surface or when the adjacent tooth overlaps the tooth being restored, preventing a lingual approach. The outline should be as conservative as possible, preserving the facial enamel (Fig 10-12).

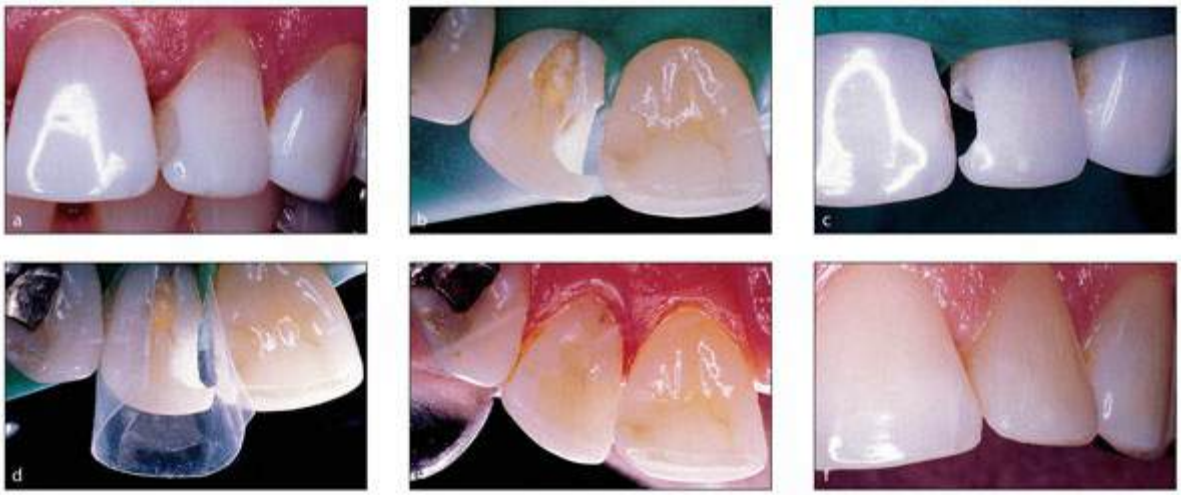


Fig 10-12 *(a)* A discolored anterior resin composite restoration to be replaced to improve esthetics. *(b)* Lingual view after the existing resin composite was removed, revealing unsupported facial enamel. *(c)* Completed preparation; observe that labial unsupported enamel has been preserved. *(d)* Clear matrix and interproximal wedge in place. *(e)* Lingual view of the finished and polished resin composite restoration. *(f)* Facial view of the Class 3 restoration illustrates the improved esthetic outcome.

Bevels have been advocated by some authors to reduce enamel fracture and provide a gradual shade transition for esthetics.[96,97] However, enamel bonds have been demonstrated clinically to be adequate without bevels.[13,98] An in vitro study using a silver-nitrate tracer revealed no significant differences in marginal microleakage in Class 3 restorations whether or not margins were beveled.[99] A flame-shaped finishing carbide or fine diamond bur, a gingival margin trimmer, or another hand instrument may be used to place bevels. Although a very wide bevel usually should be avoided, in situations where maximum esthetics is required, the bevel can be extended to 2 mm or more on the facial surface. Adequate depth of this bevel allows proper overlapping of the different opacities of the restorative layers, and the increased length allows a blending transition of the resin material onto the tooth surface. If the cervical enamel will be eliminated or compromised by a bevel with a resultant margin in or near dentin, the beveling procedure should be avoided in the cervical area (see Fig 10-9).

Resin-modified glass-ionomer restorations

Because of their cariostatic effect, resin-modified glass-ionomer restorative materials may be used in Class 3 restorations in patients at high risk for caries.[62,100] Preparations for these materials should resemble those for resin composite, but bevels are not prepared. The only reason why noncarious tooth structure should be

removed is to allow access for excavation of the carious dentin. Because these materials bond to enamel and dentin, the placement of retention grooves or points is not necessary[101] (Fig 10-13).

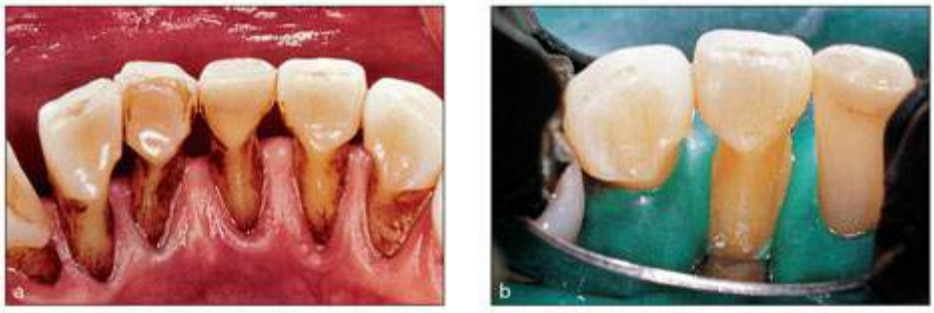


Fig 10-13*(a)* Lingual view of extensive root surface exposure and root caries lesions; the teeth are to be restored with resin-modified glass-ionomer material. *(b)* Completed excavation of carious tooth structure; the adhesive nature of the restorative material does not demand preparation of mechanical resistance and retention features. (Courtesy of Robert H. Poindexter, Kerrville, Texas.)

Class 4 interproximal lesions involving the incisal angle

Cavity preparation for interproximal lesions involving the incisal angle follows the conventional form of the Class 3 preparation and includes a portion of the incisal edge. Carious tooth structure and weak enamel are removed, and all enamel margins are beveled (Fig 10-14). A modified Class 4 preparation with extensive loss of incisal enamel is shown in Fig 10-15.

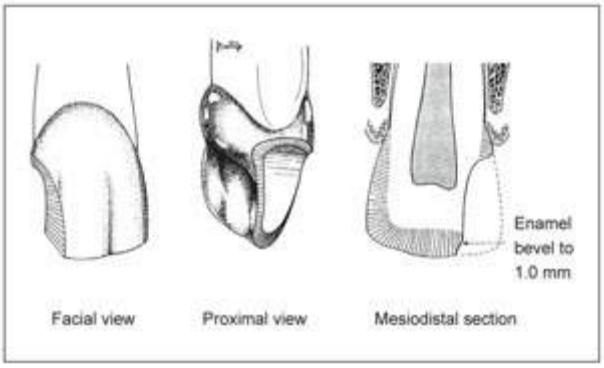


Fig 10-14 Typical Class 4 preparation. Incisal fracture caused by undermining and weakening of incisal enamel associated with a Class 3 lesion may necessitate a Class 4 cavity preparation, which is similar to a Class 3 preparation but includes a portion of the incisal edge.

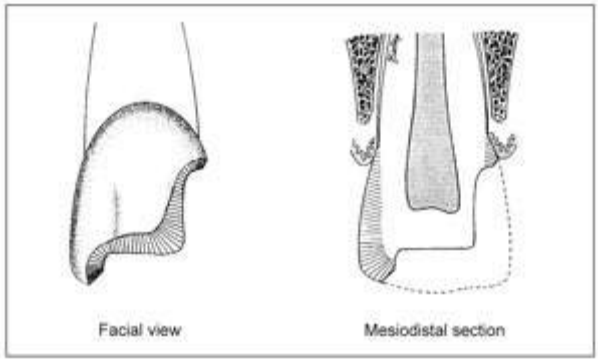


Fig 10-15 Class 4 preparation when loss of incisal enamel is extensive.

When a fracture has caused a need for restoration, if there is no carious or pulpal involvement, tooth preparation consists of rounding any sharp angles and placing a bevel on all enamel margins[102] (Fig 10-16). An enamel bevel of at least 1 mm should be placed around the periphery of the cavity. Increasing the width of the bevel beyond 1 mm has been shown to provide no additional strength,[103] but a wider bevel may provide a more harmonious esthetic blend between the resin composite and enamel. When replacing resin composite restorations, a preparation similar to that of a fractured tooth should be performed.

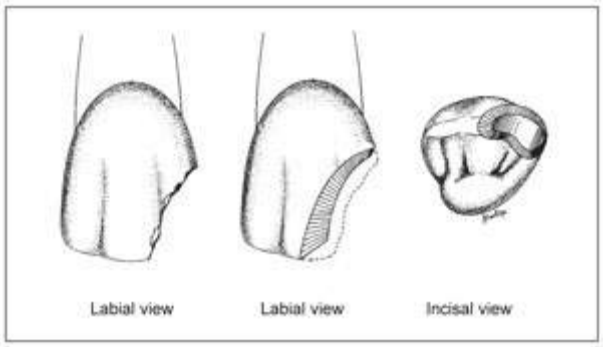


Fig 10-16 Fractures often require no preparation other than an enamel bevel.

On the facial surface, a longer bevel is needed for better esthetics. This bevel has a 60-degree angulation and is 2 to 3 mm in length. The bevel presents a scalloped or irregular out-line, has a variable thickness, starts inside the dentinoenamel junction (DEJ), and feathers and disappears onto the enamel surface (Fig 10-17). The purpose of this long bevel is to make the composite restoration blend onto the natural tooth structure and to make the transition between the two structures imperceptible. This functional-esthetic facial bevel is created first with a fine flame-shaped diamond bur and then blended on the surface with a medium-grit polishing disk. The polishing disk is also used to round sharp angles as needed. The facial bevel is often described as infinite because its margins are difficult to detect after they have been

blended with the disk and they appear to be disappearing onto the surface (Fig 10-18).

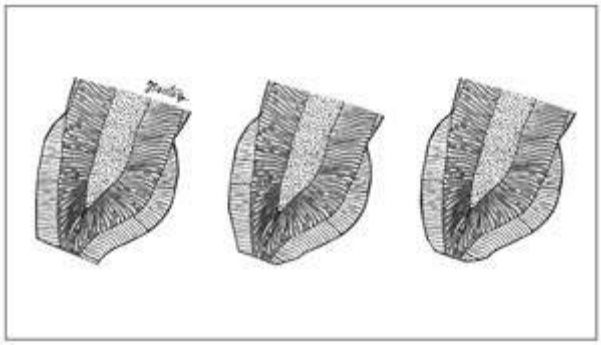


Fig 10-17 Facial and lingual bevels. The facial bevel is a disappearing bevel to allow feathering and blending of resin composite over tooth structure. Initial bevel placement is shown in the middle diagram. The facial-incisal line angle is rounded, and the facial bevel is blended to provide the disappearing bevel in the right diagram.



Fig 10-18*(a)* Existing discolored resin composite restoration in need of replacement. *(b)* Scalloped infinite bevel preparation for a Class 4 resin composite restoration.

On the lingual surface, where functional requirements are more important, the bevel should remain shorter than on the facial surface, limited to about 1 mm in length. The outline of the bevel can be straight and well defined. The bevel starts at the DEJ and has a 45-degree angulation (see Fig 10-17). In areas of strong occlusal stresses, cavity preparation must be designed to allow for sufficient thickness of resin composite so it is fracture resistant. A deeper bevel or a chamfer is prepared in areas exposed to occlusal loads in order to provide adequate marginal strength to the restoration.

If the original tooth fragment is available after traumatic fracture, in some instances the fragment may be reattached to the tooth by etching and bonding the fractured surfaces.[104–107] Clinical trials have shown these reattachments to be successful in terms of retention, in some cases for more than 7 years.[108,109] Fragment reattachment can often provide a more esthetic result than can a resin composite restoration, as long as the transition between the bonded fragment and the tooth is masked by preparing an enamel bevel and placing composite and blending it over

the enamel surfaces. The bevels should not extend into the dentin to allow the fragment to be repositioned correctly.

Use of pins

Retentive pins are not needed in resin composite restorations and should not be used. The adhesive technique provides sufficient retention, and the use of metallic pins in resin composite restorations can greatly reduce the esthetic appearance.[110] A study concluded that there was only a small (10%) increase in fracture resistance of large Class 4 resin composite restorations if pins supplemented bonding.[111] Pins in anterior teeth may encroach on the pulp and may discolor the restoration due to corrosion with marginal leakage. Some clinicians argue that if the adhesive bond is broken, it is better to lose the restoration than to have it held in place by the pins (Fig 10-19).

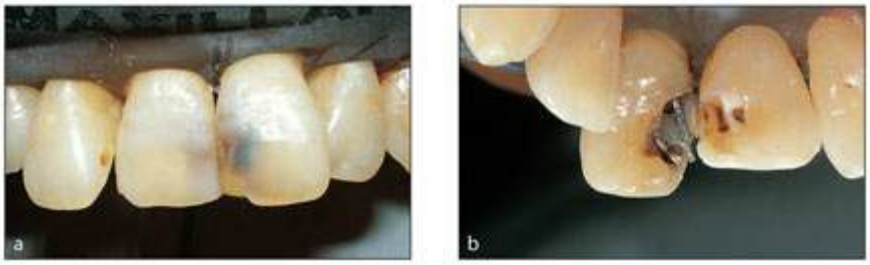


Fig 10-19*(a)* Labial view of a Class 4 resin composite restoration retained with two pins. Note the wear and color change of the composite material and evidence of corrosion discoloration or recurrent caries. *(b)* Lingual view after removal of the composite material.

Matrices

Class 3 restorations

The clear plastic strip is the most commonly used matrix with Class 3 restorations (Fig 10-20). The clear plastic matrix, when properly wedged, will reduce flash (excess material) at the gingival margin. It is placed between the teeth and adjacent to the cavity preparation. The resin composite may be shaped with an instrument, or the matrix strip may be pulled snugly around the tooth and held in place manually to provide shape to the restoration and intimately adapt the resin composite.

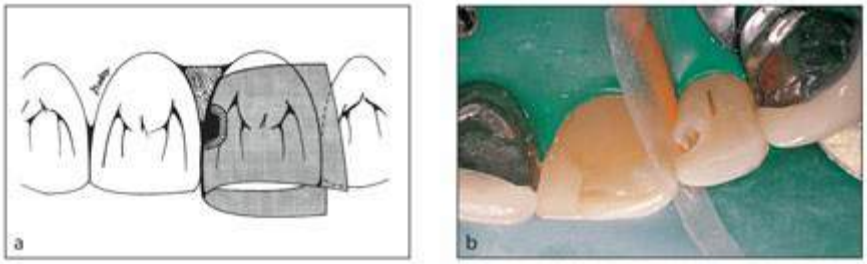


Fig 10-20*(a and b)* Lingual view of matrix and wedge placement for a typical Class 3 resin composite restoration.

Class 4 restorations

Different forms of matrices can be used to form the lingual and proximal aspects of a Class 4 restoration. One option is to use a portion of a thin, clear plastic crown form positioned to provide support until the lingual and proximal resin has been polymerized (Fig 10-21). The crown form should be trimmed to fit approximately 1 mm past the prepared margins. If the crown form is thicker than a clear plastic matrix strip, the contact areas should be thinned with an abrasive disk to allow contact of the restoration with the adjacent tooth. For Class 4 restorations with existing or mock-up resins in place, the clinician may prefer to use polyvinyl siloxane (PVS) impression material or putty to fabricate a lingual matrix that assists in shaping lingual and proximal contours (see Figs 10-25b to 10-25d). For creation of extensive restorations in which lingual contours and/or incisal length need to be changed, these may be developed in wax on a diagnostic cast of the teeth; then the lingual matrix is fabricated on the diagnostic wax-up. The matrix should extend to the facioincisal line angle to guide the shaping of the incisal edge and facilitate the creation of an incisal halo.

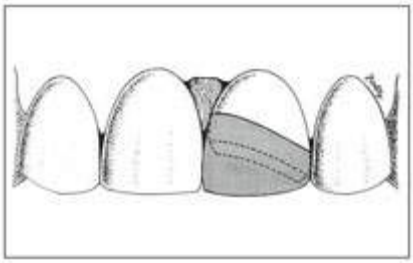


Fig 10-21 Clear plastic crown form and wedge in place for a Class 4 restoration.

Wedging

Wooden wedges are inserted between the teeth and against the matrix to seal the gingival margin, separate the teeth, protect interproximal gingiva, ensure adequate proximal contact, and push the rubber dam and proximal tissue gingivally to open the

gingival embrasure (see Fig 8-26). With proper rubber dam isolation, it is usually possible to have direct access to the gingival margin to be able to place and seal composite without the need for a wedge. Floss ligatures can greatly help in gaining access to the gingival margins by pushing the rubber dam and gingival tissues apically. When rubber dam cannot be used, care should be taken when inserting wedges because of the susceptibility of the gingiva to bleed upon removal. If the wedge will cause deformation of the matrix or poor cervical contours, it should not be used, or its use may be delayed until after a freehand increment of resin composite has been placed and sculpted in the cervical portion of the preparation and polymerized.

Placement and Curing of Resin Composite Restorations

Incremental placement and curing

Incremental placement and curing of the resin composite may be necessary for large light-cured restorations. Most resin composite should be placed in thicknesses not greater than 2 mm. In restorations exceeding 2 mm in thickness, incremental placement will ensure that each portion of composite is exposed sufficiently to the curing light and is adequately polymerized. Incremental placement is generally considered the standard of care in large preparations and is believed to offset some of the effects of polymerization stresses.[3] Although there is no definitive clinical data to support one composite placement method over another,[112] bulk curing of composite raises strong concerns over increased stress generation and tooth deformation.[29,39]

Polymerization of resin composite is initiated by the formation of free radicals that cause covalent bonding between resin monomers to produce a hardened resin composite structure. These free radicals are highly reactive to oxygen, and when they come into contact with air at the surface of the resin composite, an unpolymerized *air-inhibited layer* is formed.[113] A thin air-inhibited layer contains free radicals and is therefore reactive to new resin composite material applied to it. Even in the absence of an air-inhibited layer, if the restoration is freshly placed, free radicals remain that can induce a high degree of chemical attachment for the added increment.[114,115] Each increment should be exposed to the curing light for the appropriate material- and curing light–specific time. Inappropriately short curing times or the use of a poorly functioning curing light will result in resin composite

restorations with inferior physical properties and an increased chance that unreacted monomer, which may be cytotoxic and/or irritating to tissues, will leach out of the restoration with time.[116]

Visible light–curing units

Optimal polymerization depends on several factors: wavelength of the emitted light, *irradiance* of the light source (calculated as the power of the light divided by the area over which it is emitted; measured in mW/cm^2), light exposure time, *radiant exposure* (defined as the product of the irradiance and time; measured in mJ/cm^2), distance from light tip to composite surface, and type and shade of resin composite. Any light-curing unit with the prescribed wavelength may be used with any resin composite. However, it can be difficult to determine the optimal exposure time for any given combination of curing light and composite because recommendations on exposure time from curing light manufacturers and composite manufacturers often do not match. As a result, many clinicians will overexpose the restorations to ensure a cure as complete as possible.[117] The risk of increased exposure times, especially with high-intensity lights, is the generation of excessive heat within the tooth and in the surrounding tissues.[118,119]

Several types of light sources can be used for visible-light curing. Light-emitting diode (LED) devices have become increasingly popular in recent years because they are more energy efficient, lightweight, and portable (battery powered). The latest iteration of LEDs now generate multiple wavelengths at high irradiance and are therefore able to polymerize any type of restorative material.[120] Most dental composites contain the same photoinitiator, camphorquinone, as described earlier in this chapter. However, because camphorquinone is yellow and residual molecules that do not react may lend a yellowish color to the cured composite, manufacturers have identified less yellow–colored photoinitiators that absorb light energy at lower wavelengths (ie, closer to the UV spectrum at about 400 nm). The standard LED has minimal light output in this wavelength range, and therefore curing of resins containing these products necessitates the use of these newer multiple wavelength lights. Quartz-tungsten-halogen (QTH) lights are still widely found in dental offices and can cure effectively, as long as the incandescent bulb is well maintained and replaced periodically. Unfortunately, studies have reported very poor curing light maintenance in private practice.[121,122] Clinicians should use radiometers to assess the performance of their light units. Radiometers can be used to compare relative values of light output over time, but they may provide inaccurate values and cannot

be used to predict clinical performance.[123,124] Argon ion lasers and plasma arc curing units (PAC) are known for their very high intensity and short exposure times. However, these lights are much more costly than QTH or LED units. Furthermore, serious concerns have been raised about the stress generated in the composite when curing occurs at such a fast rate.[125]

For a more in-depth discussion of curing lights, see chapter 11.

Placement technique

Class 3 restorations

Bonding procedures should be carried out before placing the resin composite restoration. The entire preparation and a 2- to 3-mm area beyond the margins should be bonded for retention and to ensure proper marginal seal (see chapter 9 for a complete discussion of dentin adhesives) (Fig 10-22). The dentin adhesive system may be applied and polymerized either before or after the matrix and wedge have been placed (Fig 10-23). The dentist must take into consideration that inserting a well-adapted matrix can make access for etching, placing, and properly thinning the adhesive system more challenging.

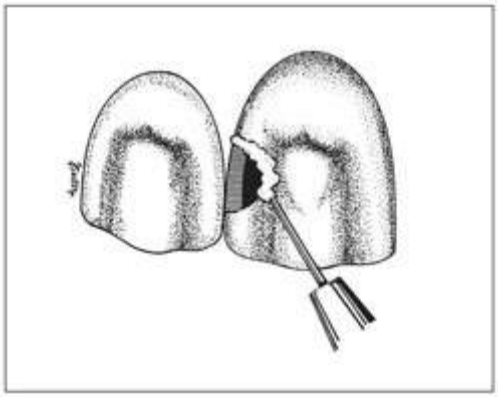


Fig 10-22 Application of gel etchant using a syringe tip. A matrix strip should be placed first to protect the adjacent tooth from the acid.

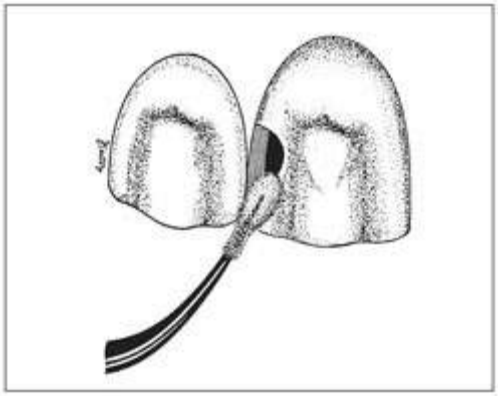


Fig 10-23 The primer and adhesive are placed with disposable brushes or applicators, and the adhesive is light cured in accordance with the manufacturer's protocol.

The resin composite dispensing tip should be inserted into the preparation and slowly injected. The composite can also be dispensed on a paper pad, rolled in a ball, and taken to the preparation with an instrument. Small Class 3 preparations of less than 2 mm can be filled in a single increment. Larger Class 3 restorations should be incrementally filled. Placement of multiple increments is recommended to optimize the degree of conversion of the resin composite in deep areas, to minimize the effects of polymerization shrinkage stress (Fig 10-24), and to create polychromatic restorations.

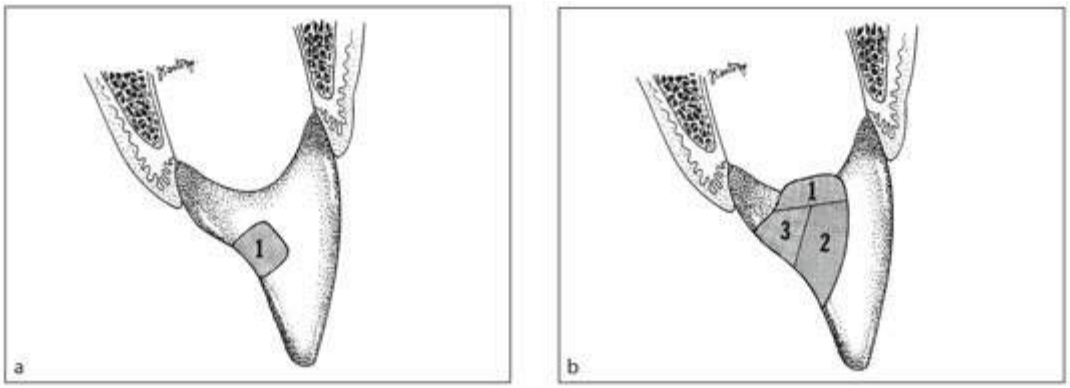


Fig 10-24*(a)* Small preparations may be filled in a single increment. *(b)* Large preparations require multiple increments to minimize the effects of polymerization shrinkage and ensure adequate polymerization.

After injection, the resin composite is adapted to the cavity walls and margins. A thin, bladed instrument is used for adaptation and contouring, or the matrix strip may be pulled tightly around the tooth to achieve close adaptation. When the material has been light cured, the wedge and matrix strip are removed and the restoration is inspected for voids. If external voids are present, they may be filled with additional resin composite material, which is then light cured.

Large “through and through” (ie, no facial or lingual tooth structure is left) Class 3 restorations often do not blend with the color and translucency of the surrounding tooth structure, and sometimes a show-through of the darkness of the mouth can be seen. In order to predictably achieve an imperceptible restoration, the *layering* or *stratified technique* should be employed. The cavity preparation is filled from the internal aspect toward the external aspect. The anatomical dentin is replaced with a dentinlike material, which is one or two shades darker than the basic shade. A 2-mm layer is placed against the axial wall and carefully adapted. One to two increments of dentinlike material may be necessary to replace the anatomical dentin. The final dentin increment should overlap part of the bevel (if present) to facilitate blending and masking. An enamel-like material, of the same shade as the basic shade, is then placed facially over the dentin increment to replace the anatomical enamel. This increment extends beyond the bevel and feathers over the natural tooth surface. This facial enamel increment should be placed in one application and built to establish the final facial contour of the restoration. Additionally, this increment should be in contact with the adjacent tooth to create the facial aspect of the proximal contact. After polymerization of the facial increment, a clear matrix can be placed interproximally to facilitate application and adaptation of the final enamel-like increment. The final increment is then placed over the lingual portion to complete the layering effect. If the clear matrix is used, it is pulled slightly toward the facial wall to carry the material and improve adaptation. Each increment of material should be light cured for approximately 20 seconds. The basic clinical steps for a Class 3 resin composite restoration are outlined in Box 10-2.

Box 10-2	**Clinical steps for a Class 3 or Class 4 resin composite restoration**

1. Select a shade before dehydration of the tooth.
2. Place a rubber dam.
3. Prewedging will aid in achieving an adequate proximal contact. Prewedge to protect the rubber dam and proximal gingival tissues during tooth preparation.
4. Initiate the cavity preparation by accessing the caries lesion through the marginal ridge with a no. 329 or 330 bur in a high-speed handpiece. Remove the proximal plate of enamel. Be careful to avoid damaging the adja cent tooth.
5. Remove the carious dentin with a round bur in a low speed handpiece running at very low speed.
6. Remove unsupported enamel, if appropriate, but pre serve as much facial enamel as possible and place bev els with a finishing bur and/or gingival margin trimmer.

7. Etch the enamel. Be careful not to etch the adjacent tooth; protect it with a matrix strip.
8. Place the primer and adhesive, following the manufac turer instructions.
9. Light cure as indicated.
10. If the preparation is large, place resin composite into the deep areas.
11. Light cure in increments no more than 2 mm thick for the appropriate time (material- and curing light–specific).
12. Place a clear plastic strip or other matrix and wedge.
13. Add composite and contour the matrix strip to contain the material in the proper shape.
14. Light cure.
15. Remove the wedge and matrix strip and inspect the restoration for voids; add resin composite if necessary.
16. Remove gingival flash with a no. 12 or 12B scalpel blade.
17. Remove flash from the other margins and contour the restoration with a finishing bur, finishing diamond, or abrasive disk.
18. Remove the rubber dam.
19. Check the occlusion and adjust as necessary.
20. Finish and polish with disks, rubber points, etc.
21. Apply etchant to the surface and margins; rinse, then apply and cure rebonding resin.

Class 4 restorations

Class 4 restorations are often complicated because of the difficulty in duplicating the natural tooth anatomical contours, surface texture, shade gradation, and opacities. To facilitate achievement of proper contours, custom matrices are employed. When treating a fracture of an incisal edge, a missing Class 4 restoration, or an existing Class 4 restoration that needs replacement, a custom matrix can be fabricated. In one technique, impression material or a rapid-set PVS material may be formed to provide a lingual matrix.[126] It should cover the lingual and incisal aspects of the tooth to be restored plus the lingual surfaces of one or two adjacent teeth. The matrix should extend facially only far enough to reach the facioincisal line angle. If tooth structure or a portion of a restoration is missing, then a quick mock-up of the restoration can be made. Expired resin composite or a seldom-used resin composite shade may be employed for this purpose. The tooth is dried, and composite is applied to form the desired tooth contours and light cured. Care should be taken in shaping the lingual aspect. After polymerization, the occlusion in maximum intercuspation and in excursive mandibular movements should be checked. The lingual contour can be finalized with a round or football-shaped carbide finishing

bur. A lingual matrix of impression material that reproduces the correct lingual contours, anterior guidance, and incisal length in the final restoration is then made.

In most cases, resin composite is layered in 2-mm increments to achieve desired coloring and complete polymerization. The incremental placement technique allows the clinician to shape the restoration to the desired form and contour. The preparation may be very slightly overfilled to allow subtractive contouring and finishing.

An *overlay technique* also may be used for Class 4 restorations to obtain a combination of strength and a very smooth surface.[127,128] The bulk of the restoration is built with a hybrid resin composite to provide strength. The final layer is a veneer of microfilled resin composite to provide a smooth, glossy surface. However, several modern hybrid and nanofilled resin composites present adequate strength and final gloss; thus, they can be used alone for the entire restoration.

To achieve imperceptible polychromatic restorations, a layering technique employing both enamel- and dentinlike materials is recommended[49,129] (Fig 10-25). Selection of dentin and enamel shades is based on the principle of the *basic shade*, in which the dentinlike material is one to two shades darker or more chromatic than the basic shade and the enamel-like material is the same shade as the basic shade. Dentin lobes, areas of high translucency, and halos may be incorporated into the restoration during layering.



Fig 10-25*(a)* Discolored, unesthetic Class 4 restoration that the patient desired to have replaced. *(b)* Lingual matrix fabricated from impression material to serve as a guide for contours and incisal length. *(c)* The defective restoration is removed, and long, scalloped bevels are placed for retention and esthetic blending of the tooth-restoration interface. *(d)* Application of the first layer of resin composite to restore lingual incisal contours. *(e)* A second layer of dentin-shade material is applied to mimic the underlying opacity and to attempt to mimic the shape of the dental mamelons. *(f)* Translucent material is applied at the incisal third to imitate incisal translucency. *(g)* An external layer of enamel-shade resin composite is feathered onto the facial surface and blended into the existing contours for a polychromatic, natural appearance. *(h)* A 12-fluted finishing bur establishes final contours and surface characteristics before employing disks and points to provide a final polish. *(i)* Flexible polishing disks conform to the facial contours of the tooth. *(j)* Abrasive-impregnated rubber points and/or cups may also be employed for polishing lingual contours and interproximal surfaces that are inaccessible to disks. *(k)* The completed restoration with a pleasing color match, surface characterization, and incisal edge morphology yields a highly esthetic outcome.

After proper etching and adhesive procedures are completed, the first layer is built up. A thin (0.2- to 0.3-mm) layer of enamel-shade composite resin is placed over the custom matrix and is limited to the size of the missing anatomical enamel

(see Fig 10-25d). The material should extend to the facioincisal line angle over the matrix. The custom matrix with the resin composite is carried to place in the mouth and positioned on the palatal aspect of the teeth. The resin composite is further adapted to the lingual bevel, if necessary, with an ovoid-shaped burnisher or a brush. The increment is light cured for the recommended time and the custom matrix removed. The second increment, the proper dentinlike material, should be placed to fill the angle produced by the surface of the first increment and the wall of tooth structure. The third increment, also a dentinlike material, reproduces the dentinal lobes (see Fig 10-25e). An interproximal carver or an ovoid-shaped bur-nisher is useful for forming these lobes, and a brush can be used to smooth the surface. When it is necessary to accentuate or make the lobes visible from the labial aspect or to create a high translucent incisal edge, a small increment of a translucent material is placed between the lobes, feathered over the incisal edge and over the bevel, smoothed, and light cured (see Fig 10-25f). A view from the incisal edge with an intraoral mirror should be used to evaluate the facial contour of each increment and to ensure that there is enough space for the following layer. A final enamel-like material increment is then placed to complete the facial contour of the restoration (see Fig 10-25g). This final layer should be placed in one increment to avoid voids and seams and contoured and shaped before polymerization until it closely resembles the desired shape. A matrix strip or an interproximal carver instrument is used to remove any excess and provide final proximal contours. Brushes are used to smooth and contour this final increment. After polymerization, the restoration is contoured, finished, and polished (see Figs 10-25h to 10-25k).

Figure 10-25 shows a step-by-step Class 4 resin composite restoration from beginning to end. The basic clinical steps for a Class 4 resin composite restoration are also outlined in Box 10-2.

Direct Resin Composite Veneers

Indirectly fabricated ceramic veneers are the gold standard for esthetics and longevity; however, direct resin composite veneers offer several advantages. Direct veneers generally provide a more conservative approach and can be placed in one visit without laboratory involvement or laboratory fees. Although the chair time required for placement of direct veneers can be considerable and experience is necessary to consistently achieve optimal results, the cost to the patient is generally lower than that for ceramic veneers. The lower cost may help to provide an alternate solution for many esthetic problems. Commonly, little or no enamel removal is

required. Resin composite will not cause premature wear of the opposing dentition and can be easily repaired and modified, unlike ceramic materials. However, resin composite veneers do not maintain their appearance as well as ceramic restorations over time. The percentage of practitioners providing direct ("freehand") resin composite veneers has declined in recent years, with a corresponding increase in the use of porcelain veneer restorations.[130] The clinical steps for a direct resin composite veneer are outlined in Box 10-3.

Box 10-3	**Clinical steps for a direct resin composite veneer**

1. Select resin shades prior to dehydration of the tooth.
2. Place a rubber dam and no. 212 (retractor) clamp, if desired. If rubber dam is not used, place gingival retraction cord to control sulcular fluid and retract the gingival tissue.
3. In most cases, the composite material is bonded directly to the tooth surface. If it is necessary to remove tooth structure to establish proper tooth alignment or to create space to mask dark tooth structure, a blunt ended diamond is recommended. Remove the smallest amount of tooth structure necessary to achieve the desired objective.
4. Etch the tooth surface with an appropriate etchant, such as 37% phosphoric acid. Protect adjacent teeth from the etchant with clear plastic strips.
5. Rinse the tooth thoroughly and dry the etched tooth surface with a stream of air.
6. Place a clear plastic strip or other matrix and wedge interproximally; apply adhesive resin and light cure for the appropriate time (material-specific instructions).
7. Apply opaque resin, if indicated, and light cure.
8. Add the selected resin composite, adapt and contour the material, and light cure for the appropriate length of time.
9. Add additional resin composite as necessary to achieve the proper shape, color, and translucency. Light cure each increment for the appropriate length of time.
10. Contour the gingival margins and remove flash with no. 12 or 12B scalpel blade.
11. Contour and finish the composite material with a carbide or diamond finishing bur.
12. Repeat the above process on adjacent teeth, if indicated.
13. Remove the rubber dam, if used.
14. Check the occlusion and adjust as necessary with finishing burs.
15. Finish and polish with disks, polishing points, etc.

16. Apply low-viscosity rebonding resin (surface sealer) to the restoration surface and margins.

Tooth preparation for direct resin veneers varies greatly, mainly according to tooth position and coloration. According to the position of the tooth, preparation may or may not be required. Teeth positioned lingually or those that are short cervicoincisally rarely require any preparation and can be brought into correct alignment and length by simply adding resin composite. Facially positioned teeth with or without rotation require some degree of preparation to allow space for the thickness of the material and to bring the tooth in alignment with the arch. Discolored teeth require preparation only if there is not enough space, about 1 mm, over the facial aspect to place a masking agent and resin composite.

Direct resin composite veneers can be produced by using single opacity shades or by layering various shades and opacities to obtain polychromatic esthetic restorations according to adjacent teeth, desired results, and patient expectations (Fig 10-26). If a tooth shows a dark discoloration, an opaque layer or masking agent is placed over the adhesive and polymerized[131] (Figs 10-27 and 10-28a to 10-28d). When necessary, white, orange, or brown tints may be used to add a natural look to the cervical or proximal surfaces of teeth or to internally characterize a restoration (Figs 10-28e and 10-28f). These color modifications will influence the surface shade unless the thickness of the resin composite exceeds 2.5 mm.[88–90] Translucency, the appearance of lobes or mamelons, and/or white or bluish characterization of the incisal edge may be added in the incisal third of the composite veneer. An attempt should be made to sculpt the resin composite to desired contours. If the restoration is slightly overcontoured, it may be finished and polished to proper contours. If it is slightly undercontoured in any area, additional resin composite material may be added.

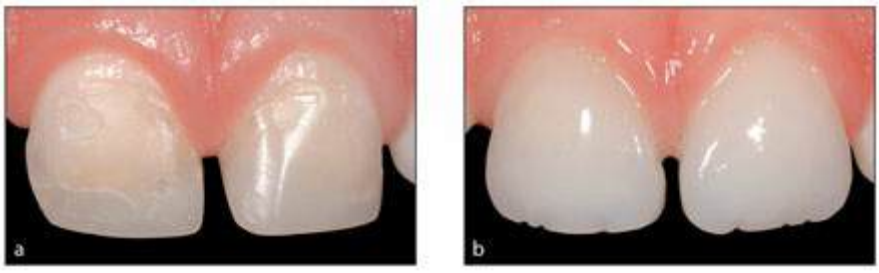


Fig 10-26*(a)* Maxillary central incisors exhibit external stain, hypocalcification, and discolored restorations that require replacement. *(b)* Completed direct veneers mask stains and meet esthetic demands in this young patient.

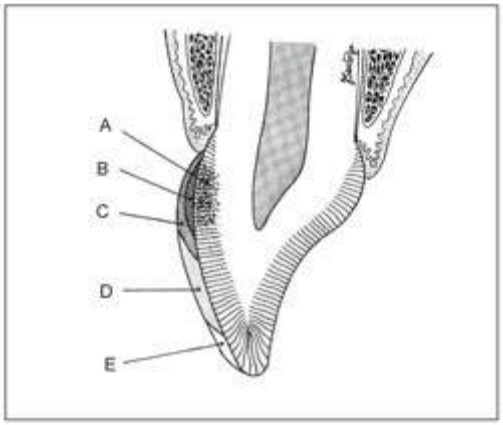


Fig 10-27 Facial veneering and masking of heavy stain/fluorosis using a combination of opaque, cervical, body, and translucent shades: stain/fluorosis (A); opaque shade (B); cervical shade (C); body shade (D); translucent shade (E).



Fig 10-28*(a)* Dark composite restorations on maxillary central incisors need replacement. *(b)* Isolation is achieved with a rubber dam. For veneer preparation and restoration, it is necessary to obtain maximum cervical retraction with floss ligatures. *(c)* The right central incisor has been prepared for a Class 4 restoration. The left central incisor has been prepared for a complete facial veneer. An intrinsic discoloration is present on the left central incisor, which will require masking with a layer of opaque composite. *(d)* Completed facial veneer on the left central incisor and Class 4 restoration on the right central incisor match the color of the adjacent teeth. A pleasing esthetic result with effective masking of stains can be achieved when the correct composite opacity is selected. When needed, stains can be used to internally characterize restorations. *(e)* Internal characterization with white and caramel stains are applied to the resin composite. *(f)* Final result showing lifelike restorations with internal characterization.

Creating a diagnostic wax-up is recommended to reduce clinic time and increase the predictability of the final veneers. The wax-up will provide information on the final contours and the amount of material necessary to complete the veneers. The

diagnostic wax-up can be shown to the patients for their acceptance, or an impression can be made by PVS and an intraoral mock-up performed[132] (Fig 10-29). A custom matrix made from the wax-up, similar to the one described for a Class 4 restoration, can be fabricated from the wax-up to aid with placement of the veneers.

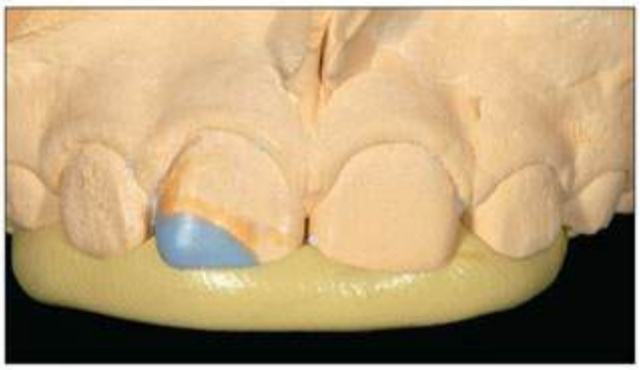


Fig 10-29 Wax-up of a Class 4 restoration for diagnostic purposes and patient presentation. Additionally, a PVS matrix can be made from the wax-up to aid in the restorative phase.

Diastema closure

The technique for diastema closure is similar to that for placement of direct veneers and Class 4 restorations. In most cases, no tooth structure has to be removed, and the resin composite is retained solely by adhesive bonding. For a small diastema (1 mm or less), an enamel-like resin composite is added to the proximal surfaces of the two adjacent teeth (Fig 10-30). For a moderate-sized to large diastema, an opaque or dentinlike material should be placed to block the show-through effect caused by the darkness of the mouth (Fig 10-31). A slight blending of the material onto the facial surface will help achieve a natural transition of shades and improve the esthetic outcome. The facial and incisal contours can then be established with an enamel-like resin composite. When closing a diastema, care should be taken to provide the proper tooth contours when placing and finishing the material. This is especially crucial in the gingival embrasure areas. If the diastema exceeds 2.5 mm, it may be necessary to use a combination of direct veneering and orthodontic movement to position the teeth into a more easily managed and esthetically pleasing location. Corrective enameloplasty of the distal-surface contours of the teeth to be restored, followed by building up of adjacent teeth, may improve the esthetic outcome.



Fig 10-30 *(a)* A maxillary lateral incisor and canine with an existing diastema to be closed with resin composite. *(b)* Phosphoric acid etchant is applied; no preparation of natural tooth structure is required for adhesion. *(c)* Completed diastema closure showing improved esthetics and imperceptible blend of resin composite and natural enamel. Note the translucency of the incisal edge.



Fig 10-31 *(a)* Labial view of finished diastema closure showing the use of body and translucent shades to simulate natural tooth color and translucency. *(b)* Moderate midline diastema. The patient's desire for esthetic improvement may be satisfied with resin composite. *(c)* Completed diastema closure. Note the physiologic contouring of the cervical aspect of the restorations and the homogenous luster of the composite material and natural tooth surface. (Figs 10-31b and 10-31c courtesy of Robert H. Poindexter, Kerrville, Texas.)

When diastema closure is performed, occlusal relationships and esthetic proportions as well as the overall facial esthetics must be considered. When anterior teeth are widened, it may also be necessary to lengthen them to preserve natural anatomical proportions. If occlusal relationships and facial appearance will allow, the proper tooth length can be established by adding to the incisal edge. It is also possible to improve the length-to-width ratio by surgical crown lengthening in some patients. The desired lengths and widths of teeth should be determined using a diagnostic wax-up before treatment is begun (Fig 10-32). A trial application or mock-up, assessing the esthetic alteration of the shape and color of the proposed restoration, may be accomplished with resin composite applied to unetched teeth.[134]



Fig 10-32 Diagnostic wax-up of proposed recontouring and closure of dia-stemata. This treatment planning procedure may assist in identifying tooth-size discrepancies, areas requiring esthetic recontouring, and gingival height and contour considerations. The cast may also be used for patient education and communication of treatment goals, as well as for fabrication of a lingual matrix and as a chairside guide for direct restorations.

Maintenance of the proper length and width relationships in anterior teeth is very important to achieving an esthetic result for resin composite veneers, porcelain veneers, and diastema closures (see chapter 3). A study evaluating the length and width of anterior teeth revealed that, on average, central incisors and canines are approximately equal in length and are 20% longer than lateral incisors.[133] The proper dimensions, length-to-width ratios, and width-to-width ratios are shown in Tables 10-1 and 10-2.

Table 10-1 **Dimensional averages (in mm) for maxillary incisors and canines***

	Males		Females	
Tooth	**Length**	**Width**	**Length**	**Width**
Central incisor	10.7	9.4	9.6	9.1
Lateral incisor	9.1	7.5	8.2	7.1
Canine	10.7	8.5	9.2	8.0

*Data from Gillen et al.[133]

Table 10-2 **Length/width and width/width* ratios for maxillary incisors and canines†**

	Males		
	Length/ width	**Width/width lateral**	**Width/width canine**
Central incisor	1.15:1	1.25:1	1.1:1
Lateral incisor	1.2:1	–	
Canine	1.25:1	1.15:1	–
	Females		
	Length/ width	**Width/width lateral**	**Width/width canine**
Central incisor	1.05:1	1.3:1	1.15:1
Lateral incisor	1.15:1	–	

Canine	1.15:1	1.15:1	–

*Width/width ratios are only listed for larger teeth to smaller teeth (eg, central incisor to lateral incisor).

†Data from Gillen et al.[133]

A periodontal probe or caliper may be used as a measuring device to evenly divide the space to be closed. Diastema closures frequently require augmenting two to six teeth with resin composite to achieve optimal esthetic relationships (Figs 10-33 and 10-34).



Fig 10-33*(a and b)* Diastema closures frequently require bonding of resin composite to two to six teeth to achieve proper esthetic relationships. Multiple composite veneers were needed for this patient.

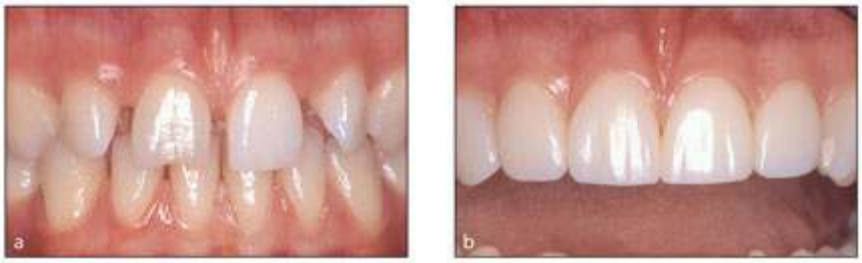


Fig 10-34*(a and b)* This diastema closure also required multiple composite veneers.

Most diastema closures can be accomplished without the use of a matrix or a wedge, thus providing better control of proximal contours in the gingival embrasure areas. When restoring adjacent teeth, the proximal contours are best built up one at a time. Sculptable microfilled or nanofilled resin composites are the preferred materials. A small amount of resin composite, just enough to provide half of the diastema closure, is placed on the mesiofacial line angle of the first tooth. A thin, bladed instrument is used to push and adapt the material on the facial surface. The remaining composite is carried toward the lingual, and the gingival embrasure is

contoured to the shape desired. Care should be taken to achieve the desired proximal contour and tooth width. It is very helpful to measure the width of the restored tooth with calipers and compare it to the adjacent tooth and the remaining space to ensure that proper symmetry is achieved. Following polymerization, the proximal surface of the first restoration should be finished and polished to the exact contour desired. The contours of the facial surface and incisal edge should be very close to the final desired shape but may be polished at the same time as the adjacent restoration. Fig 10-35 shows a step-by-step diastema closure without the use of a matrix.



Fig 10-35 Diastema closure using resin composite. *(a)* The diastema between maxillary central incisors measures slightly more than 1 mm. *(b)* Resin composite is placed into the diastema and polymerized to verify shade selection prior to the dehydration of the teeth that accompanies the

procedure. *(c)* Tooth surfaces are cleaned with a slurry of fine pumice. *(d)* Retraction cord has been placed to move the tissue of the interproximal gingival papilla. *(e)* Etchant is applied to only one central incisor; the etched area is larger than that needed for attachment of resin composite. A strip of clear, plastic matrix protects the other central incisor from being etched. After 30 seconds of etching, the surface is rinsed and thoroughly dried. *(f)* Adhesive resin is placed over the etched enamel and light polymerized. *(g)* Resin composite is placed so that it has slightly greater contour than the desired final restoration. *(h)* The resin composite has been light polymerized. *(i)* The resin composite is contoured with a coarse-grit disk. *(j)* The interproximal area is finished so that the resin composite will close one-half of the diastema. *(k)* Final polishing of the restoration is performed. *(l)* The restoration of the left central incisor is complete. *(m)* The mesial portion of the right central incisor is etched; a plastic matrix strip protects the adjacent tooth from the etchant. *(n)* After the etchant is rinsed away, the surface dried, and the resin adhesive applied and polymerized, resin composite is placed in contact with the adjacent restoration and polymerized. *(o)* The restorations are lightly separated by pushing a thin instrument against the contact from the gingival embrasure.



Fig 10-35*(cont)* *(p)* The gingival margins of the restoration are shaped with a no. 12 scalpel blade. *(q)* The proximal surface, gingival to the interproximal contact, is finished and polished using progressively finer-grit finishing strips. *(r)* Surfaces of the restorations are polished with an aluminum oxide polishing paste. *(s)* The diastema closure procedure is complete. (Courtesy of J. William Robbins, San Antonio, Texas.)

A tight, properly contoured contact can be achieved using a “pull-through” matrix technique. When placing the adjacent restoration, care should be taken to confine both the etchant and the bonding resin to prevent inadvertent splinting. After the adhesive resin is polymerized, a clear celluloid strip is placed in the contact area and the resin composite applied to the facial surface of the second tooth. After the material is blended to contour and shaped against the strip, the strip is slowly pulled to the lingual, drawing the material with it. When the material is midway through the contact, the strip should be removed with a tug, leaving the material in the contact

area. Wiping the strip with alcohol-dampened gauze prior to placement may help to release the material from the strip. A brush or placement instrument can be used to gently push the facial resin increment into contact with the already completed restoration on the adjacent tooth, and then the embrasures are shaped and refined and the composite polymerized. Although the polymerized resin will stick to the adjacent tooth, it can easily be separated by lightly rotating (torquing) an instrument between the teeth. Following separation, a clear celluloid strip is reinserted and contoured against the lingual contour of the tooth. A small amount of resin composite is placed against the strip to fill any deficiencies on the lingual surface, and the lingual portion of the strip is drawn tightly against the cingulum with a thumb or a finger. The strip is then pulled out to the facial, drawing the material into the lingual deficiencies. After polymerization, the teeth are again separated and finishing procedures are initiated.

Finishing and Polishing

Finishing includes the shaping, contouring, and smoothing of the restoration, while *polishing* imparts the shine or luster to the surface (see Fig 10-25k). Finishing a restoration is as important as adequate shade matching. Even when a good shade match has been obtained, if the finishing process does not simulate the adjacent tooth contours, the restoration will not appropriately blend with its surroundings. There are many products available for finishing and polishing, including diamond and carbide burs, scalpel blades and carving instruments, various types of flexible disks, abrasive-impregnated rubber points and cups, metal and plastic finishing strips, and polishing pastes.[134] The smoothest possible surface is obtained when the resin composite polymerizes against a clear plastic strip without subsequent finishing or polishing. However, such a surface typically has a higher resin content, because the monomers tend to work their way through the fillers like a sieve when the composite is pushed against a surface. This may yield a very smooth superficial layer but one that is less resistant to wear due to its reduced amount of reinforcing fillers.

The finishing and polishing process can affect many aspects of the final restoration, including surface shine, surface staining, and plaque accumulation. The composition of the composite material and the polishing system and protocol used all influence the finish and polish that can be obtained.[135–138] Polish retention over time is an important concern for clinicians and depends greatly on filler particle size. Composites with larger particles tend to be more difficult for the clinician to obtain smooth surfaces, and they show more surface roughness with time, as the

surface is exposed to abrasion and erosion from food and drinks.[139,141]

Instruments

Diamond and carbide burs

Ideally, a minimum of finishing should be imparted to a resin composite. This goal is easily achieved when contouring and smoothing of the restoration is obtained before polymerization. The 12-fluted carbide burs have traditionally been used to perform gross finishing of resin composites. Fine finishing diamonds are also available for finishing resin composite restorations and have been found to impart less surface damage to microfilled resin composites than carbide finishing burs[141] (Fig 10-36). These finishing burs and diamonds may be used to develop the proper macro- and microanatomy for the restoration. The transition from resin to enamel should be slowly smoothed until it is undetectable. These burs can be used dry to better visualize the margins and anatomy being developed, and they should always be used with light pressure to avoid overheating and possibly damaging the resin composite surface.

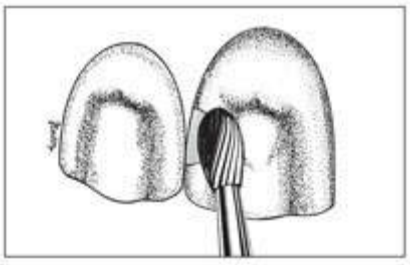


Fig 10-36 A finishing bur or fine diamond is used for gross finishing.

Scalpel blades and resin carving instruments

Blades and carving instruments are effective at refining gingival margins and interproximal areas. Burs and disks can cause damage to the soft tissue and cementum in the gingival area. A no. 12 or 12B scalpel blade is often used to safely remove excess material at the gingival margin. Blades and resin carving instruments are also used to refine embrasures (Fig 10-37).

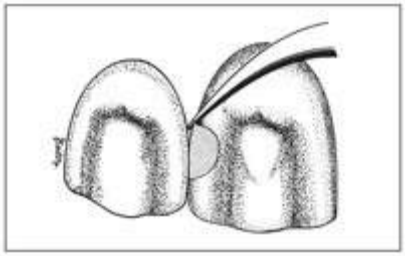


Fig 10-37 Gingival flash is removed with a no. 12 or 12B scalpel blade.

Disks

One brand of flexible disks (Sof-Lex, 3M ESPE) has practically become the standard in finishing and polishing. The disks in one Sof-Lex series have a soft, flexible backing and a series of grits that can provide a smooth, even finish. Another Sof-Lex series, and similar disks made by other manufacturers, have thin plastic or polymeric backings that allow access of the abrasive side into embrasures and interproximal areas. When all four grits are used in sequence, these flexible finishing disks are reported to provide either the smoothest or nearly the smoothest surfaces of any finishing systems.[136,142,143] Sequential use of disks with progressively finer grits produces a smooth, durable finish (Fig 10-38). Unfortunately, disks have the tendency to leave flat surfaces, unlike the normal rounded shapes found in natural teeth.

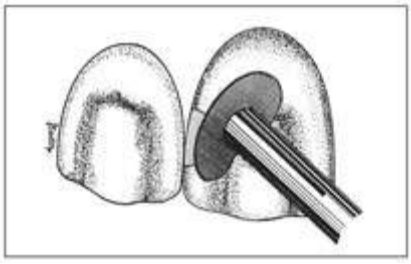


Fig 10-38 Polishing of the final restoration may be accomplished with flexible abrasive disks or abrasive-impregnated rubber points, cups, and/ or disks.

Dry finishing with disks used in sequence is reported to be superior or equal to wet finishing for smoothness, hardness, and color stability.[144] However, dry finishing tends to clog disks with abrasive particles and makes the disks work less efficiently.

Impregnated rubber points and cups

A wide variety of rubber finishing and polishing points and cups impregnated with abrasive materials are available. Like disks, rubber cups and points are used

sequentially from coarse to fine grit. The coarse grits may be effective for gross reduction and finishing, while the fine grits create a smooth, shiny surface. The primary advantage of rubber points and cups over disks is that they provide access to grooves, desirable surface irregularities, and the concave lingual surfaces of anterior teeth. The surfaces obtained with these cups tends to better resemble the natural anatomy of teeth.

Finishing strips

Finishing strips are used to contour and polish the proximal surfaces and margins gingival to the interproximal contact (Fig 10-39). They are available with metal or plastic backings. Most metal-backed strips are used for gross reduction, but care must be taken not to overreduce the restoration; these metal-backed strips will also remove enamel, cementum, and dentin. Plastic strips come in various widths and grits and can be used for both finishing and polishing. Like the flexible disks, finishing strips come in a series of grits, which should be used in series from coarsest to finest.

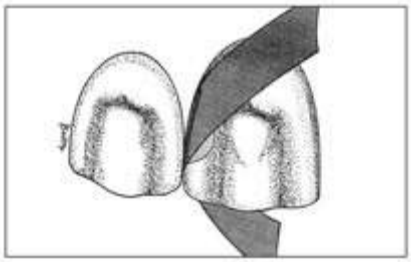


Fig 10-39 Polishing strips are used to contour and polish the proximal surface and margin.

Procedures

Contouring

Contouring has the purpose of shaping the restoration to simulate the natural contours and anatomy of a tooth. It is extremely important for the clinician to be familiar with the anatomy found in the natural dentition to readily and predictably obtain good restoration contours.

Contouring should be carried out in a sequence. The first step in contouring is the evaluation of the length of the restoration in relation with the adjacent teeth and the establishment of the facioincisal line angle. A medium-grit disk is used to reduce and contour the incisal edge. Young patients will have round incisal line angles in a mesiodistal direction because of the lack of wear; however, with age the teeth

become flatter as a result of wear. A palatal inclination of the incisal edge is observed with age in the maxillary anterior teeth. In contrast, a labial inclination is observed for the mandibular anterior teeth. The second step is to check the occlusion in maximum intercuspation and in excursive movements and adjust accordingly. The third step is to locate and contour the mesiofacial and distofacial transitional line angles. This is accomplished by using fine carbide or diamond burs or impregnated rubber cups. The transitional line angles are easily identified from an incisal view with an intraoral mirror. The fourth step is to contour tooth embrasures. The incisal embrasures can be easily contoured with medium-grit disks and the gingival embrasures with a no. 12 or 12B scalpel blade. Additionally, a perforated diamond strip (eg, Visionflex, Brasseler) is very effective in smoothing and contouring the gingival embrasures.

Finishing and characterization

Finishing should be done carefully to avoid damaging the surface or margins of the resin composite restoration. Medium-and fine-grit impregnated cups are sequentially used to progressively smooth the facial and lingual surfaces of the restoration. Similarly, a sequence of finishing strips is used on the proximal surfaces and margins. The dental floss should pass over the proximal resin surface smoothly and snap through the contact without shredding.

Surface characterization is obtained by observing and copying details of the adjacent dentition onto the restoration. A fine diamond at low speed is useful to reproduce imbrication lines found in young teeth.

Polishing

A final surface polish can be obtained by using a diamond or silicon carbide–impregnated disk or cup. Aluminum oxide or diamond polishing pastes are also used to obtain a high polish.

Many operators have observed the development of a “white line” at the margins of resin composite restorations during finishing. The exact cause of this phenomenon is not known, but several investigators and clinicians have put forward possible explanations. One explanation[145–147] implicates traumatic finishing leading to microfractures in the resin composite or tooth structure at the interface. Other proposed causes include improperly rotating abrasive disks (the disks should rotate from resin composite to tooth),[148,149] inadequate polymerization of the resin composite material,[147] and polymerization shrinkage causing microfracture of

unsupported or fragile enamel at the margins.[150] When the white line presents an esthetic problem and more conservative procedures such as rebonding do not resolve it, the white area must be removed with a bur and additional composite must be bonded and finished.

A typical kit for finishing resin composites contains a fine diamond and a mandrel for disks to contour the restoration (Fig 10-40). It also contains rubber points to sequentially smooth the restoration and to provide final luster. Figure 10-41 illustrates a contouring and polishing sequence.



Fig 10-40 Kit for finishing and polishing resin composites.



Fig 10-41*(a)* Preoperative view of chipped resin composite restoration in need of replacement. *(b)* Cavity preparation for a Class 4 resin composite and facial veneering. *(c)* The first step in contouring is to establish the length of the restoration and position of the facioincisal line angle. *(d)* The position of the mesio- and distofacial line angles are marked with a pencil for visualization. *(e)* A fine flame-shaped diamond is used to contour the restoration and reproduce the fine anatomical details. *(f)* Surface characterization of an extracted human incisor. Some natural dentition is characterized by abundant secondary anatomy. *(g)* A coarse-grit silicon-impregnated rubber cup, the green Jiffy Polisher (Ultradent), is used to smooth the roughness left by the diamond bur. *(h)* A medium-grit rubber cup, Gloss Plus Polishers (Kerr), is used to further smooth the restoration. *(i)* A fine-grit diamond-impregnated cup, HiLuster Plus (Kerr), is used to obtain high gloss. *(j)* Final gloss is obtained before rubber dam removal. *(k)* The definitive restoration matches the surface characterization, gloss, and contours of the adjacent teeth.

Rebonding

Rebonding (also called *surface sealing* or *glazing*) is performed after the restoration is finished and polished. The margins are re-etched, and a coat of

unfilled or lightly filled low-viscosity resin is placed over the restoration surface and polymerized. Rebonding has been reported to improve marginal integrity, aid color stability, improve early wear resistance, and help reduce staining of the restoration.[151–156] The long-term effect of rebonding is questionable because this surface layer of low-viscosity resin is likely to wear quickly. A number of low-viscosity resins, called *surface sealers*, are now available for use in the rebonding procedure. Rebonding is rarely needed for anterior restorations when a proper placement and finishing protocol has been followed.

Techniques for Repair or Correction of Intraoral Restorations

Resin composites are effective for the repair or correction of intraoral restorations for many kinds of materials. By combining chemical bonding techniques as well as macro- and micromechanical retention, attachment of the resin composite to the old restoration can be obtained. In many cases, repair or correction requires less preparation and will reduce the risk of damage to the tooth when compared with complete replacement of defective or unesthetic restorations.[157,158] Macromechanical retention can be created by the preparation of undercuts in the old restoration, which can also improve the resistance form.

Micromechanical retention can be obtained by preparation with a coarse diamond bur or air abrasion. Air abrasion is effective on all resin composites, porcelain, acrylic resins, and metals. It may be performed with aluminum oxide particles or with a special aluminum oxide powder coated with a layer of silica (CoJet Sand, 3M ESPE).[159–161] The latter is designed to deposit a layer of silica on the surface of the air-abraded restoration. This may enhance chemical retention and allow silanization. On porcelain and resin composites containing glass, silanization is an effective method to maximize the chemical bond. Some silanes are directly applied to the surface, while others are activated when mixed with specific adhesive primers or resins. On metals, specific metal primers, applied after air abrasion, are effective in improving the bond strength.[162]

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